

Central Visayan Institute Foundation

It is the home and pioneer of the prominent school-based innovation known as the Dynamic Learning Program (DLP). The DLP is a synthesis of classical and modern pedagogical theories adapted to foster the highest level of learning, creativity, and productivity. The school takes pride in its Research Center for Theoretical Physics (RCTP) established in 1992, which organizes small international workshops to foster the informal but intense exchange of ideas and perspectives on outstanding problems in physics and mathematics.

Lesson 4: Indigenous Science and Technology in the Philippines

Indigenous Knowledge System

Indigenous knowledge is embedded in the daily life experiences of young children as they grow up. They live and grow in a society where the members of the community prominently practice indigenous knowledge. Their parents and other older folks served as their first teachers and their methods of teaching are very effective in transmitting cultural knowledge in their minds. The lessons they learned are intimately interwoven with their culture and the environment. These lessons comprised of good values and life stories of people on their daily life struggles. Their views about nature and their reflections on their experiences in daily life are evident in their stories, poems, and songs.

Some examples of indigenous knowledge that are taught and practiced by the indigenous people are:

- predicting weather conditions and seasons using knowledge in observing animals' behavior and celestial bodies
- using herbal medicine;
- preserving foods;
- classifying plants and animals into families and groups based on cultural properties;
- preserving and selecting good seeds for planting;
- using indigenous technology in daily lives;
- building local irrigation systems;
- classifying different types of soil for planting based on cultural properties;
- producing wines and juices from tropical fruits; and
- keeping the custom of growing plants and vegetables in the yard.

Indigenous Science

Indigenous science is part of the indigenous knowledge system practiced by different groups of people and early civilizations (Gribbin, 2001; Mkapa, 2004; Sibisi, 2004). It includes complex arrays of knowledge, expertise, practices, and representations that guide human societies in their enumerable interactions with the natural milieu: agriculture, medicine, naming and explaining natural phenomena, and strategies for coping with changing environments (Pawilen, 2005). Ogawa (1995) claimed that it is collectively lived in and experienced by the people of a given culture.

According to Cajete (2004), indigenous science includes everything, practiced by indigenous peoples both past and present. Iaccarino (2003) elaborated this idea by explaining that science is a part of culture, and how science is done largely depends on the cultural practices of the people.

Indigenous beliefs also develop desirable values that are relevant or consistent to scientific attitudes as identified by Johnston (2000), namely: (1) **motivating attitudes**; (2) **cooperating attitudes**; (3) **practical attitudes**; and (4) **reflective attitudes**. These cultural beliefs therefore can be good foundation for developing positive values toward learning and doing science and in bringing science in a personal level.

Pawilen (2005) explained that indigenous science knowledge has developed diverse structures and contents through the interplay between the society and the environment. According to Kuhn (1962), developmental stages of most sciences are characterized by continual competition between a number of distinct views of nature, each partially derived from, and all roughly compatible with the dictates of scientific observation and method. Sibisi (2004) also pointed out that indigenous science provides the basics of astronomy, pharmacology, food technology, or metallurgy, which were derived from traditional knowledge and practices.

Pawilen (2006) developed a simple framework for understanding indigenous science. Accordingly, indigenous science is composed of traditional knowledge that uses science process skills and guided by community values and culture.

1. Indigenous science uses science process skills such as observing, comparing, classifying, measuring, problem solving, inferring, communicating, and predicting.
2. Indigenous science is guided by culture and community values such as the following:
 - The land is a source of life. It is a precious gift from the creator.
 - The Earth is revered as "Mother Earth." It is the origin of their identity as people

- All living and nonliving things are interconnected and interdependent with each other.
 - Human beings are stewards or trustee of the land and other natural resources. They have a responsibility to preserve it.
 - Nature is a friend to human beings-it needs respect and proper care.
3. Indigenous science is composed of traditional knowledge practiced and valued by people and communities such as ethno-biology, ethno-medicine, indigenous farming methods, and folk astronomy.

Indigenous science is important in the development of science and technology in the Philippines. Like the ancient civilizations, indigenous science gave birth to the development of science and technology as a field and as a discipline. Indigenous science helped the people in understanding the natural environment and in coping with everyday life. UNESCO's Declaration on Science and the Use of Scientific Knowledge (1999) recognized indigenous science as a historical and valuable contribution to science and technology.

SEMIFINAL

CHAPTER II: SCIENCE, TECHNOLOGY AND SOCIETY AND THE HUMAN CONDITION

Lesson 1: Human Flourishing

Eudaimonia, literally "good spirited," is a term coined by renowned Greek philosopher Aristotle (385-323 BC) to describe the pinnacle of happiness that is attainable by humans. This has often been translated into "human flourishing" in literature, arguably likening humans to flowers achieving their full bloom. As discussed in the Nicomachean Ethics, Aristotle's human flourishing arises as a result of different components such as phronesis, friendship, wealth, and power. In the Ancient Greek society, they believe that acquiring these qualities will surely bring the seekers happiness, which in effect allows them to partake in the greater notion of what we call the Good.

As times change, elements that comprise human flourishing changed, which are subject to the dynamic social history as written by humans. People found means to live more comfortably, explore more places, develop more products, and make more money, and then repeating the process in full circle. In the beginning, early people relied on simple machines to make hunting and gathering easier. This development allowed them to make grander and more sophisticated machines to aid them in their endeavours that eventually led to space explorations, medicine innovations, and ventures of life after death. Our concept of human flourishing today proves to be different from what Aristotle originally perceived then- humans of today are expected to become a "man of the world." He is supposed to situate himself in a global neighbourhood, working side by side among institutions and the government to be able to reach a common goal. Competition as a means of survival has become passé; coordination is the new trend.

Interestingly, there exists a discrepancy between eastern and western conception regarding society and human flourishing. It has been observed that western civilization tends to be more focused on the individual, while those from the east are more community-centric. Human flourishing as an end then is primarily more of a concern for western civilizations over eastern ones. This is not to discredit our kinsfolk from the east; perhaps in their view, community takes the highest regard that the individual should sacrifice himself for the sake of the society. This is apparent in the Chinese Confucian system or the Japanese Bushido, both of which view the whole as greater than their components. The Chinese and the Japanese encourage studies of literature, sciences, and art, not entirely for oneself but in service of a greater cause. The Greek Aristotelian view, on the other hand, aims for eudaimonia as the ultimate good; there is no indication whatsoever that Aristotle entailed it instrumental to achieve some other goals. Perhaps, a

person who has achieved such state would want to serve the community, but that is brought upon through deliberation based on his values rather than his belief that the state is greater than him, and thus is only appropriate that he should recognize it as a higher entity worthy of service.

Nevertheless, such stereotypes cannot be said to be true given the current stance of globalization. Flourishing borders allowed people full access to cultures that as a result, very few are able to maintain their original philosophies. It is in this regard that we would tackle human flourishing--in a global perspective and as a man of the world.

Science, Technology, and Human Flourishing

In the previous chapters, contributions of science and technology have been laid down thoroughly. Every discovery, innovation, and success contributes to our pool of human knowledge. Perhaps, one of the most prevalent themes is human's perpetual need to locate him in the world by finding proofs to trace evolution. The business of uncovering the secrets of the universe answers the question of our existence and provides us something to look forward to. Having a particular role, which is uniquely ours, elicits our idea of self-importance. It is in this regard that human flourishing is deeply intertwined with goal setting relevant to science and technology. In this case, the latter is relevant as a tool in achieving the former or echoing Heideggers statement, technology is a human activity that we excel in as a result of achieving science. Suffice to say that the end goals of both science and technology and human flourishing are related in that the good is inherently related to the truth. The following are two concepts about science which ventures its claim on truth.

Science as Method and Results

For the most part, science's reputation stems from the objectivity brought upon by an arbitrary, rigid methodology whose very character absolves it from any accusation of prejudice. Such infamy effectively raised science in a pedestal untouchable by other institutions-its sole claim to reason and empiricism garnering supporters who want to defend it and its ways.

In school, the scientific method is introduced in the earlier part of discussions. Even though the number of steps varies, it presents a general idea of how to do science:

1. Observe and determine if there are unexplained occurrences unfolding.
2. Determine the problem and identify factors involved.
3. Through past knowledge of similar instance, formulate hypothesis that could explain the said phenomenon. Ideally, the goal is to reject the null hypothesis and accept the

alternative hypothesis for the study "to count as significant" (can also be separated into additional steps such as "to generate prediction" or "to infer from past experiments").

4. Conduct experiment by setting up dependent and independent variables, and trying to see how independent ones affect dependent ones.
5. Gather and analyze results throughout and upon culmination of the experiment. Examine if the data gathered are significant enough to conclude results.
6. Formulate conclusion and provide recommendation in case others would want to broaden the study.

At least in the students' formative years, the above routine is basic methodology when introducing them to experimentation and empiricism two distinct features that give science edge over other schools of thought. Throughout the course of history, however, there exists heavy objections on the scientific procedure; the line separating science and the so-called pseudoscience becomes more muddled.

Verification Theory

The earliest criterion that distinguishes philosophy and science is verification theory. The idea proposes that a discipline is science if it can be confirmed or interpreted in the event of an alternative hypothesis being accepted. In that regard, said theory gives premium to empiricism and only takes into account those results which are measurable and experiments which are repeatable. This was espoused by a movement in the early twentieth century called the Vienna Circle, a group of scholars who believed that only those which can be observed should be regarded as meaningful and reject those which cannot be directly accessed as meaningless. Initially, this proved to be attractive due to general consensus from people, which happened to see for themselves how the experiment occurred, solidifying its validity and garnering supporters from esteemed figures. Its shortcomings, however, proved to be a somewhat too risky- several budding theories that lack empirical results might be shot down prematurely, causing slower innovation and punishing ingenuity of newer, novel thoughts. Celebrated discoveries in physics, for instance, are initially theorized without proper acknowledgment of their being. Einstein's theory on the existence of gravitational waves would, following this thought, be dismissed due to lack of evidence almost a hundred years ago. (Quantum mechanics would not have prospered if the scientific society during the time of Edwin Schrödinger did not entertain his outrageous thought that the cat in the box is both dead and alive, which can only be determined once you look in the box yourself.

Aside from above critique, this theory completely fails to weed bogus arguments that explain things coincidentally. A classic example is astrology, whose followers are able to

employ the verification method in ascertaining its reliability. The idea is that since one already has some sort of expectations on what to find, they will interpret events in line with said expectations, American philosopher Thomas Kuhn warned us against bridging the gap between evidence and theory by attempting to interpret the former according to our own biases, that is, whether or not we subscribe to the theory. Below is a short story illustrating this point:

↗ **Suppose, for instance, this girl, Lea has a (not-so- scientific) theory that her classmate Ian likes her. Good, she thought, I like him too. But how do I know that he likes me?**

She began by observing him and his interactions with her. Several gestures she noted include his always exchanging pleasantries with her whenever they bump into each other, his big smile when he sees her, and him going out of his way to greet her even when riding a jeepney. Through these observations, she was then able to conclude that Ian does like her because, she thought, **why would anyone do something like that for a person he does not like?**

As it turns out, however, Ian is just generally happy to meet people he knew. He had known Lea since they were in first year and regards her as a generally okay person. It is no surprise then that upon learning that Ian basically does this to everyone, Lea was crushed. She vowed to herself that she would never assume again. Based from above story, is it justified for Lea to think that Ian does not like her? Not quite. The next criterion also warns us about the danger of this view.

Falsification Theory

Perhaps the current prevalent methodology in science, falsification theory asserts that as long as an ideology is not proven to be false and can best explain a phenomenon over alternative theories, we should accept the said ideology. Due to its hospitable character, the shift to this theory allowed emergence of theories otherwise rejected by verification theory. It does not promote ultimate adoption of one theory but instead encourages research in order to determine which among the theories can stand the test of falsification. The strongest one is that which is able to remain upheld amidst various tests, while being able to make particularly risky predictions about the world. Karl Popper is the known proponent of this view. He was notorious for stating that up-and-coming theories of the time, such as Marx's Theory of Social History and Sigmund Freud's Psychoanalysis, are not testable and thus not falsifiable, and subsequently questioning their status as scientific. Albeit majority of scientists nowadays are more inclined to be Popperian in their beliefs, this theory, similar to the theory above, presents certain dangers by interpreting otherwise independent evidence in light of their pet theory.

To illustrate, previous story is restated:

Ian is generally everybody's friend. He likes to be around people and generally aspires to become everybody's friend. However, there is this one girl, Lea, who seemed to not like him when he is around. Every time he waves at her, she turns away, and when they are in the same room, she avoids his glances. Through this, he concluded that Lea does not like him and does his best to show her that he is not a threat. He began greeting her whenever they pass by each other at the corridor, even going so far as calling her attention when he was in the jeepney and saw her walking past. When they are able to talk to each other, he found out that Lea is just really shy and is not accustomed to people greeting her. He then was able to conclude that his initial impression of her not liking him (as a person) is wrong and thus said proposition is rejected.

Although there is no happy ending yet for Lea and Ian, we can thus see how in this case, falsification method is prone to the same generalizations committed by the verification method. There is no known rule as to the number of instance that a theory is rejected or falsified in order for it to be set aside. Similarly, there is no assurance that observable event or "evidences" are indeed manifestations of a certain concept or "theories." Thus, even though, theoretically, falsification method is more accepted, scientists are still not convinced that it should be regarded as what makes a discipline scientific.

Science as a Social Endeavor

Due to inconclusiveness of the methodologies previously cited, a new school of thought on the proper demarcation criterion of science emerged. Several philosophers such as Paul Thagard, Imre Lakatos, Helen Longino, David Bloor, and Richard Rorty, among others, presented an alternative demarcation that explores the social dimension of science and effectively, technology. Sciences cease to belong solely to gown-wearing, bespectacled scientists at laboratories. The new view perpetuates a dimension which generally benefits the society. For instance, far-off places in South America where many of the tribes remain uncontacted, do not regard western science as their science. Whatever their science is, it can be ascertained that it is in no way inferior to that of globalized peoples' science. Thus, it presents an alternative notion that goes beyond the boundaries of cold, hard facts of science and instead projects it in a different light, such as a manifestation of shared experience forging solidarity over communities.

Science and Results

For the most part, people who do not understand science are won over when the discipline is able to produce results. Similar to when Jesus performed miracles and garnered followers, people are sold over the capacity of science to do stuff they cannot

fully comprehend. In this particular argument, however, science is not the only discipline which is able to produce results-religion, luck, and human randomness are some of its contemporaries in the field. For some communities without access to science, they can turn to divination and superstition and still get the same results. Science is not entirely foolproof, such that it is correct 100% of the time. Weather reports, for one, illustrate fallibility and limitations of their scope, as well as their inability to predict disasters. The best that can be done during an upcoming disaster is to reinforce materials to be more calamity proof and restore the area upon impact. It can be then concluded that science does not monopolize the claim for definite results.

Science as Education

Aforementioned discussion notes that there is no such thing as a singular scientific method, offering instead a variety of procedures that scientists can experiment with to get results and call them science. Discoveries in physics, specifically in quantum mechanics, appeared to have debunked the idea of objectivity in reality, subscribing instead to alternative idea called intersubjectivity. With objectivity gone, it has lost its number one credence. Nevertheless, there still exists a repressing concept that comes about as a result of unjustified irreverence of science-our preference of science-inclined students over those which are less adept.

There are distinct portions in entrance exams in the secondary and tertiary levels that are dedicated to science and mathematics. In the Philippines, a large distribution of science high schools can be found all over the country, forging competition for aspiring students to secure a slot and undergo rigorous science and mathematics training based on specialized curricula. Although arguable as these schools also take great consideration in providing holistic education by assuring that other non-science courses are covered, adeptness in science and mathematics are the primary condition to be admitted. This preference is also reflected on the amount of STEM (Science, Technology, Engineering, Mathematics)-offering schools accommodating Grades 11 and 12. Among all the clusters being offered, STEM trumps the remaining clusters in terms of popularity and distribution, with Accounting and Business coming in as a close second. One might infer that there are more demand in this field as students are preconditioned that the field would latter land them high-paying jobs and a lucrative career after graduation.

How is science perceived by those who graduated from this field? A couple of years ago, a student entered a class all curious and excited. When he was made to report on Paul Feyerabend's work *How to Defend Society Against Science* one day, he looked dissident, staunchly refusing to consider the author's ideas on science and critiquing him instead. When asked why, he reasoned out that he had come from a science high school and was trained to regard science in a distinct accord. As isolated a case as it may seem, it

somewhat suggests that the aforementioned kind of academic environment has made students unwelcoming of objections against science. Reminiscent of Paul Feyerabend's sentiment above, he muses how the educational system can hone and preserve students' capacity to entertain other options and decide for themselves the best among all presented. It will thus reinforce their imagination and allow some level of unorthodoxy, bringing forth novel discoveries that otherwise would not be considered had they stuck to the default methodology. Innovations are brought forth by the visionaries, not the prude legalists, and several notable figures in science even consider themselves as outsiders.

If one is really in pursuit of human flourishing, it would make sense for them to pursue it holistically. Simply mastering science and technology would be inadequate if we are to, say, socialize with people or ruminate on our inner self. Aristotle's eudaimonic person is required to be knowledgeable about science, among other things of equal importance. They are supposed to possess intellectual virtues that will enable them to determine truth from falsehood or good reasoning from poor reasoning. A true eudaimon recognizes that flourishing requires one to excel in various dimensions, such as linguistic, kinetic, artistic, and socio-civic. Thus, he understands that he should not focus on one aspect alone.

How Much Is Too Much? In 2000, world leaders signed the Millennium Development Goals (MDG) that targets eight concerns, one of which states that they should be able to forge a global partnership for development. Inasmuch as the institutes imposing them do so in good faith, the primary goal to achieve growth for all might prove to be fatal in the long run.

Economists believe that growth is the primary indicator of development, as both go hand in hand, and has put forth their resources in trying to achieve such. Technology has been a primary instrument in enabling them to pursue said goal, utilizing resources, machineries, and labor. What is missing in this equation is that growth presents an illusory notion of sustainability-the world's resources can only provide so much, it cannot be expected to stretch out for everybody's consumption Over a long period of time. Moreover, growth is not infinite-there is no preordained ceiling once the ball starts rolling. If the MDG convention's intent was to get everyone in the growth ship, that ship will surely sink before leaving the port. The same analogy applies to the capacity of nature to accommodate us, whom Joseph Hickel contemplated on, suggesting that developed countries should not push forth more growth but instead adopt "dedevelopment" policies or else, everybody loses. The rapid pace of technological growth allows no room for nature to recuperate, resulting in exploitation and irreversible damages to nature. Right now, we are experiencing repercussions of said exploits in the hands of man-made climate change, which would snowball and affect majority of flora and fauna, driving half of the latter extinct in less than a hundred year from now. If this continues in its currently alarming rate, we might bring about our own extinction.

Lesson 2: Technology as a way of Revealing

The Human Condition Before Common Era

Our early ancestors' primal need to survive paved way for the invention of several developments. Gifted with brains more advanced than other creatures, humans are able to utilize abundant materials for their own ease and comfort. As it is difficult to pinpoint the particular period where technology is said to have started, one can say that at the very least, the motivation to make things easier has been around since humans are. **Homo erectus** have been using fire to cook, through chipping one flint over the other to produce a spark, all the while without realizing the laws of friction and heat. Tools from stone and flints marked the era of the Stone Age, during the advent of our very own **Homo sapiens**, and humans began to sharpen stones as one would a knife; an example of this is the simple machine called wedge.

This particular period proved to be difficult for our ancestors, but in a remarkably distinct way. There is little to no written accounts except for several cave drawings and unearthed artefacts from various parts of the world that narrate how their culture came to be. It then proves that there is little capacity for our ancestors to contemplate and perceive things outside themselves in a more reflective manner. For instance, there are several excavations in different parts of Europe of miniature statues prevalent during the Paleolithic period, the so-called "Venus" figure. It depicts a rudimentary carving of a voluptuous woman out of ivory or stone. The reason behind this is still unknown to archaeologists and anthropologists alike; they can only infer that the humans of yesterday have a certain fixation on the female anatomy. What did this fascination translate to? Lack of conclusive evidence hinders us from proceeding any further.

Soon enough, people discovered minerals and began forging metalwork. They realized that these substances are more durable, malleable, and have more luster than the previous material. On the other hand, there are some indicators that humans in the past share the same concerns and interests, suggesting that these inclinations persist through multitudes of generation over several millennia. Fur clothing and animal skin are primarily used for comfort against harsh winds-our ancestors are able to draw the connection between their being naked and vulnerable due to some lack of fur or protective covering which would otherwise allow them to withstand extreme weather conditions. They begin to cover themselves up out of necessity, and gradually, added several more to their garments. Some of those found at excavation sites are reminiscent of early accessories, suggesting that our ancestors have been deeply engaged in the concept of beauty.

Perhaps, they had taken a liking to a certain shiny stone, or a perfectly shaped bone, and wanted to wear it as trophy. What does this tell of them and their philosophies? Were they perhaps proud to show off their hunt and how good of a hunter they were? Were they concerned with social standing and stratification? How about the meaning of life? Were they also curious on finding explanations to certain phenomena?

At least for the last question, it seems that they have found their answer in the person of religion. Excavations on the latter half of the Stone Age include several figures thought to be ceremonial, meaning, that perhaps people of the time had also painstakingly wrought and hewed said figures in honor of some deity. This notion, as it was then and as it is now, is often people's resort to make sense of events happening outside their control. The initial roster of primitive gods includes objects they encounter through their day-to-day lives, so it is not surprising that different tribes may have different gods. Those who might have lived alongside majestic creatures, such as elephants and mammoths, might have been awed by their size and worshiped them as the owner of the land, asking for blessings in their hunting ground. On the contrary, they might have hunted the mammoths for their woolly coat and meat, taking down the animal for the entire community to eat. In windy places near mountains, they might have had a mountain god to explain wind currents and ask for provisions. On the other hand, those who were near coastal areas or bodies of water might have had water gods they referred to when asking for a good catch. However, it might be also the case that people of prior civilizations shared several generic gods, such as the sun.

Nevertheless, it can be positively inferred that like the people of today, our ancestors also found the need to explain things in a way that makes sense to them. They quickly realized that there are events outside of their control and attempted to justify things as being a work of a supernatural being. Throughout the course of history, religion remains to be the strongest contender to science arguably due to its being the most easily grasped. Admittedly, once people stop connecting the dots between cause and effect, they turn to something that could possibly explain their inadequacies in making sense of the world. The people of yesterday appeared to have acknowledged early on that they could only do and understand as much, that perhaps other powers at play also existed alongside them. This notion effectively humbled and perhaps grounded them, with their constant befuddlement serving as an early reminder that they were way behind several larger, more powerful forces in nature in terms of order of things.

The Human Condition in the Common Era

For a long time, humans were content with their relationship with nature. Earliest case of man-made extinction occurred over 12,000 years ago, possibly brought upon by hunting and territorial disputes. The Holocene extinction, also called the sixth extinction or more aptly Anthropocene extinction, occurred from as early as between 100,000 to 200,000 years up to the present. It pertains to the on-going extinction of several species-both flora and fauna-due to human activity. Driven by their primal need to survive and gaining the upper hand in terms of development and adaptability, humans were quick to find ways to drive off other megafaunas threatening a prospective hunting spot and eventually settling grounds. Growing population also necessitated finding additional resources, leading to overhunting and overfishing of common prey, some of which were endemic to the area. Hunting, coupled with a changing terrain that the humans began cultivating when agriculture emerged some 9,000 years ago, caused several species to lose competition in territory and food resources. Formation of communities caused humans to expand more in territory and more people to feed; large, separate communities hailing from the same ancestors and residing in the same large community paved the way for civilizations. Even as a community, the people realized that though they were at most self-sufficient, they were in constant need of resources. Albeit waging wars with other tribes seemed to be the early solution, they were able to find out some 5,000 years ago that engaging in a peaceful negotiation was also a possible and less bloody method. They realized that they could get hold of things not present in their towns by offering something of same value present in theirs. It is in this process that trade emerged, leading to cross-town and eventually cross-cultural interaction as more products were exchanged and the initial needs extended to wants.

People then had a new objective-gather as much products as possible. They have turned to wealth as one of their goals as humans and ultimately as civilizations, for they perceived that those who have many, live comfortably and thus are generally happier than those who do not have sufficient wealth. Thus, they began to hunt, farm, and produce things with prospect of profit. A nuclear community which is initially self-sufficient has to accommodate their growing population with depleting resources, leading them to be reliant to other communities' produce which keeps them surviving. In return, these communities have to make use of their current resources twice as much to provide for other communities' needs. Products of every kind were exchanged, ranging from necessary ones such as crops, cattle, poultry, others of kind, and clothing materials, up to metals, accessories, weapons, spices, literature, and entertainment. They were able to find and create niches for interests. When they could not sell products, they used their skills and got compensated for it-bringing forth a specialized group of artisans. Humanity

became more complex. The primary goal was not merely to survive, but to live the good life.

Technology has been instrumental in all of these because in searching for the good life, people were able to come up with creations that would make life easier, more comfortable, and more enriching. Although the good life envisioned before might be pale in comparison to the multifariousness of today, it offered us the initial intricacies of how today came to be. Such intricacies are also evident in the machines created and causes endeavored by the people of long ago. They perceive death as, at the very least, unpleasant and concocted potions to ward evil off from their kinsmen, often appealing to their gods for blessings. Medicine was thus born, although it would take a considerably long time before it part ways with potion. They became fixated with gold and were adamant in procuring more, trying to use incantations with mixtures of substances to turn lead into one. This ultimately paved way for the emergence of chemistry in its primitive form, not quite distinct from alchemy. Due to differing races, belief, or abundance of resources and/or territory, wars were always being waged, leading communities to allocate resources to the militia. Initially, the early leaders were those who portrayed exceptional strength among their group-this condition carried on for generations. Physical strength was valued at most, although there appeared to be as many intellectually gifted figures just the same. These innovators were primarily the ones behind discoveries and triumphs of these civilizations.

Position-wise, the humans of today are much better off compared to humans several centuries ago. Advancements in medicine, technology, health, and education ushered in humanity's best yet, and show no sign of stopping. Below are some of the notable comparisons then and now:

1. **Mortality Rate.** Due to technology, lesser women and children die during birth, assuring a robust population and strong workforce. Medical care for premature infants allows them to survive and develop normally, while proper maternal care ensures that mothers can fully recover and remain empowered.
2. **Average Lifespan.** Aside from the reason that people engage less in combat and are less likely to die in treatable diseases now as opposed to then, science is able to prolong lives by enhancing living status and discovering different remedies to most diseases. Distribution of medicines is also made easier and faster.
3. **Literacy Rate.** Access to education provided to more individuals generally creates a more informed public that could determine a more just society.
4. **Gross Domestic Product (GDP).** Although not an indicator of an average person's lifestyle in a certain country, it is often used to determine the value of the

country's goods and services produced within the territory given a certain time period. Higher country income is brought upon by high productivity, often an indicator of presence of technology.

The Essence of Technology

Humanity has indeed come a long way from our primitive ways, and as a general rule, it is said that we are more "developed" than we were before. Above data are few indicators of the route that we have come to take as species, and there are no signs of stopping. Modern humans are reliant on technology in their search for the good life. We see ways and means from nature to utilize and achieve growth—a goal that we believe would bring forth betterment.

In retrospect, this view of technology proves to be goal-oriented. It assumes that it is instrumental in achieving a goal in mind, that it is a purposeful, deliberate craft humans steer in order to reach some greater good. In the advent of postmodernism, however, the deterministic view appended to technology crumbled as people began to question if anything is deterministic at all. Apart from its purpose, what is technology? Was the history of technology brought purposeful choices for man in his search for the ultimate good? Some tried to redefine technology away from its purpose. One philosopher by the name of Martin Heidegger argued that its essence, or purpose, and being are different from each other. He was able to expound on this point upon identifying that technology can either be perceived as first, a means to achieve man's end and second, that which constitutes human activity. The second perspective paints technology in such a way that each period reveals a particular character regarding man's being. A characteristic design, or flaw, unfolds based on the repercussions brought upon by immersing ourselves with a piece of new technology. In effect, through technology, a myriad of new questions begins to mount. Rather than thinking that humans have a clear idea of what to expect in a good life, it can be stated that technology allows humans to confront the unknown and see how they would react.

This is not a good thing altogether though, for technological revelation is but one of the many ways to perceive the world. However, as long as humans are invested in growth and development, we cannot distance ourselves from this perspective. In the name of growth, we view the world as a field of resources, vent on attributing monetary value on seemingly priceless entities. We begin to categorize nature as renewable and nonrenewable instead of seeing it as it is. Humans are reduced into the amount of productivity they are able to render during their lifetime, and our current mindset is geared toward which would utilize our own skills. A good life is one which is practical in essence; a life which makes use of our labor and which we get compensated fairly upon. It is no wonder that the sciences are one of the most sought after courses, for the opportunities are plenty and the resources

are bountiful. Since humans appear not to really know what they are seeking for, the search continues. It is a looming fear, however, that the path we are treading will not take us to the right direction, leading us in endless circles instead in our pursuit of the good life. This is the danger presented by too much reliance on technology. Humans lose track of things that matter, reducing their surroundings to their economic value. As this presents strong backing by the sciences whose reverence is also brought upon by our deluded enchantment with technology, it will prove to be a herculean task to distance ourselves from this perspective and consider alternatives. After all, it was science and technology that gave us explanations, which worked for us and benefited us. Rejecting a working, tried-and-tested process seems foolish, more so, knowing that there are no options of equal value. It will be absurd to venture the dark and the unknown, but it should be done in order for us to retrace our steps to be able to achieve the Good.

Backtracking the Human Condition

Technology's initial promises proved to be true, regardless of its ramifications. All in all the human condition improved, only if by improving we measure the level of comfort, various scientific breakthroughs, and improved lifestyles of those who had the luxury to afford to do so. Different machineries aid in prolonging lives-assisting those with disabilities, honing efficiency in industrial workplaces, and even exploring the universe for places we can thrive once all the Earth's resources are depleted. As to the initial aims, it appears that things really did not much differ. Some places in the world are still battling for their daily survival-diseases, tribe wars, lack of habitable territories, and competitions on resources are several factors contributing to such. People still wage wars on the basis of races, belief, and abundance of resources and/or territory; except that now, they are able to inflict such in a global scale. A lot of people still subscribe to religion in explaining things that they do not know. For those who have ceased to do so, they have turned their worships to reverence of science. Whether science or religion, these people are still bent on trying to make Sense of the events happening in the world on the basis of either of these two paradigms. They are still trying to discover and rediscover things that would give meaning to their lives-whether it is honor, strength, or merit. People are still trying to make sense of their existence in the world, and technology does little to aid them in their pursuit of life's meaning.

It seems that the human condition, although more sophisticated, is nothing but a rehashed version of its former self. Nothing much has changed since then, and it appears that nothing will change in the times to come if we fail to shift our view elsewhere. While it is true that technology offered us one compelling notion of the truth and the good, we should be staunch in our resolve if we want to know the real one. For starters, we might begin with considering other concepts, which corresponds to the Good, such as Aristotle's conception of human flourishing. His notion entertains the idea

of holistic enrichment of a person situated in his society. A notable distinction on Aristotle's idea is his subscription on evaluative concepts called virtues and their role in achieving the good life. Technological advancements are seemingly occurring in a rapid pace that our morality cannot quite keep up; no such consideration was given in this approach in achieving the good life. This will further be discussed in the following chapters.

Lesson 3: The Good Life

Aristotle and How We All Aspire for a Good Life

It is interesting to note that the first philosopher who approached the problem of reality from a "scientific" lens as we know now, is also the first thinker who dabbled into the complex problematization of the end goal of life: **happiness**. This man is none other than Aristotle.

Compared to his teacher and predecessor, Plato, Aristotle embarked on a different approach in figuring out reality. In contrast to Plato who thought that things in this world are not real and are only copies of the real in the **world of forms**, Aristotle puts everything back to the ground in claiming that this world is all there is to it and that this world is the only reality we can all access. For Plato, change is so perplexing that it can only make sense if there are two realities: the world of forms and the world of matter. Consider the human person. When you try to see yourself in front of the mirror, you normally say and think that you are looking at yourself—that is, you are the person who slept last night and you are the same person looking at yourself now, despite the occasional changes like a new pimple that grows on your nose. The same is true for a seed that you threw out of the garden last month. When you peek into the same patch of land where the seed ingrained itself into, you may be surprised to see a little plant showing itself to you and to the sun. Plato recognized change as a process and as a phenomenon that happens in the world, that in fact, it is constant. However, Plato also claims that despite the reality of change, things remain and they retain their ultimate "whatness"; that you remain to be you despite the pimple that now sits atop your nose. Plato was convinced that reality is full of these seemingly contrasting manifestations of change and permanence. For Plato, this can only be explained by postulating two aspects of reality, two worlds if you wish: the world of forms and the world of matter. In the world of matter, things are changing and impermanent. In the world of forms, the entities are only copies of the ideal and the models, and the forms are the only real entities. Things are red in this world because they participate in what it means to be red in the world of forms.

Aristotle, for his part, disagreed with his teacher's position and forwarded the idea that there is no reality over and above what the senses can perceive. As such, it is only by observation of the external world that one can truly understand what reality is all about. Change is a process that is inherent in things. We, along with all other entities in the world, start as potentialities and move toward actualities. The movement, of course, entails change. Consider a seed that eventually germinates and grows into a plant.

The seed that turned to become the plant underwent change- from the potential plant that is the seed to its full actuality, the plant. Aristotle extends this analysis from the external world into the province of the human person and declares that even human beings are potentialities who aspire for their actuality. Every human being moves according to some end. Every action that emanates from a human person is a function of the purpose (**telos**) that the person has. When a boy asks for a burger from a Filipino burger joint, the action that he takes is motivated primarily by the purpose that he has, inferably to get full or to taste the burger that he only sees on TV. When a girl tries to finish her degree in the university, despite the initial failures she may have had, she definitely is being propelled by a higher purpose than to just graduate. She wants something more, maybe to have a license and land a promising job in the future. Every human person, according to Aristotle, aspires for an end. This end, we have learned from the previous chapters, is happiness or human flourishing. No individual-young or old, fat or skinny, male or female-resists happiness. We all want to be happy. Aristotle claims that happiness is the De all and ends all of everything that we do. We may not realize it but the goal of everything that we do is happiness. If you ask one person why he is doing what he is doing, he may not readily say that it is happiness that motivates him. Hard-pressed to explain why he is motivated by what motivates him will reveal that happiness is the grand, motivating force in everything that he does. When Aristotle claims that we want to be happy, he does not necessarily mean the everyday happiness that we obtain when we win a competition or we eat our favorite dish in a restaurant. What Aristotle actually means is human flourishing; a kind of contentment in knowing that one is getting the best out of life. A kind of feeling that one has maxed out his potentials in the world, that he has attained the crux of his humanity.

Happiness as the Goal of a Good Life

In the eighteenth century, John Stuart Mill declared the Greatest Happiness Principle by saying that an action is right as far as it maximizes the attainment of happiness for the greatest number of people. At a time when people were skeptical about claims on the metaphysical, people could not make sense of the human flourishing that Aristotle talked about in the days of old. Mill said that individual happiness of each individual should be prioritized and collectively dictates the kind of action that should be endorsed. Consider the pronouncements against mining. When an action benefits the greatest number of

people, said action is deemed ethical. Does mining benefit rather than hurt the majority? Does it offer more benefits rather than disadvantages? Does mining result in more people getting happy rather than sad? If the answers to the said questions are in the affirmative, then the said action, mining, is deemed ethical.

The ethical is, of course, meant to lead us to the good and happy life. Through the ages, as has been expounded in the previous chapters, man has constantly struggled with the external world in order to reach human flourishing. History has given birth to different schools of thought, all of which aim for the good and happy life.

Materialism

The first materialists were the atomists in Ancient, Greece. Democritus and Leucippus led a school whose primary belief is that the world is made up of and is controlled by the tiny indivisible units in the world called **atomos or seeds**. For Democritus and his disciples, the world, including human beings, is made up of matter. There is no need to posit immaterial entities as sources of purpose. **Atomos** simply comes together randomly to form the things in the world. As such, only material entities matter. In terms of human flourishing, matter is what makes us attain happiness. We see this at work with most people who are clinging on to material wealth as the primary source of the meaning of their existence.

Hedonism

The hedonists, for their part, see the end goal of life in acquiring pleasure. Pleasure has always been the priority of hedonists. For them, life is about obtaining and indulging in pleasure because life is limited. The mantra of this school of thought is the famous, "Eat, drink, and be merry for tomorrow we die." Led by Epicurus, this school of thought also does not buy any notion of afterlife just like the materialists.

Stoicism

Another school of thought led by Epicurus, the stoics espoused the idea that to generate happiness, one must learn to distance oneself and be apathetic. The original term, *apatheia*, precisely means to be indifferent. For the stoics, happiness can only be attained by a careful practice of apathy. We should, in this worldview, adopt the fact that some things are not within our control. The sooner we realize this, the happier we can become.

Theism

Most people find the meaning of their lives using God as a fulcrum of their existence. The Philippines, as predominantly Catholic country, is witness to how people base their life goals on beliefs that hinged on some form of supernatural reality called heaven. The

ultimate basis of happiness for theists is the communion with God. The world where we are in is only just a temporary reality where we have to maneuver around while waiting for the ultimate return to the hands of God.

Humanism

Humanism as another school of thought espouses the freedom of man to carve his own destiny and to legislate his own laws, free from the shackles of a God that monitors and controls. For humanists, man is literally the captain of his own ship. Inspired by the enlightenment in seventeenth century, humanists see themselves not merely as stewards of the creation but as individuals who are in control of themselves and the world outside them. This is the spirit of most scientists who thought that the world is a place and space for freely unearthing the world in seeking for ways on how to improve the lives of its inhabitants. As a result of the motivation of the humanist current, scientists eventually turned to technology in order to ease the difficulty of life as illustrated in the previous lessons. Scientists of today meanwhile are ready to confront more sophisticated attempts at altering the world for the benefit of humanity. Some people now are willing to tamper with time and space in the name of technology. Social media, as an example, has been so far a very effective way of employing technology in purging time and space. Not very long ago, communication between two people from two continents in the planet will involve months of waiting for a mail to arrive. Seeing each other real time while talking was virtually impossible. Now, communication between two people wherever they are, is not just possible but easy. The Internet and smart phones made real-time communication possible not just between two people, but even with multiple people simultaneously.

Technology allowed us to tinker with our sexuality. Biologically male individuals can now undergo medical operation if they so wish for sexual reassignment. Breast implants are now available and can be done with relative convenience if anyone wishes to have one. Hormones may also be injected in order to alter the sexual chemicals in the body.

Whether or not we agree with these technological advancements, these are all undertaken in the hopes of attaining the good life. The balance, however, between the good life, ethics, and technology has to be attained.

Lesson 4: When Technology and Humanity Cross